

FARIA SAMANTA AKBAR

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EDUCATION

University of Toronto

Toronto, ON

B.A.Sc., Electrical and Computer Engineering, PEY Co-op

Jun 2029

- Coursework: Algorithms and Data Structures, Software Design and Communication, Software Engineering, Computer Organization, Digital Systems, Applied Deep Learning, Intro to Machine Learning.

EXPERIENCE

Orix Geoscience

Toronto, ON

Machine Learning and Computer Vision Intern

Jun 2026 – Present

- Built a Python and OpenCV pipeline as the sole engineer, converting scanned geological maps into structured, GIS-ready vector data end to end, tested on 1,500 real maps supplied by Orix.
- On a given map, the pipeline detects features at about 68% accuracy in 3 to 4 minutes, versus a minimum of 10 hours and up to 2 weeks for a person doing the same tracing manually.
- Reached 80% or higher accuracy on deeply coloured maps and about 60% on black and white maps using a LAB colour space heuristic, then trained SE-UNet and Mamba-style deep learning models in PyTorch that pushed per-map accuracy to about 98%.
- Built a Python validation tool comparing pipeline output against source maps to confirm every layer was captured and flag misaligned or dropped features, cutting repeated validation effort about 30%.
- Managed the codebase in Azure DevOps Repos, used Pipelines and Test Plans as part of the development workflow, and trained deep learning models on a rented Azure GPU-enabled virtual machine.

GoBolt

Toronto, ON

Automated Cross-Belt Sortation System, university design course, real client

Jan 2025 – Apr 2025

- Identified the automation problem, chose the design approach, and directed a 6-person team, personally programming the Python control logic that routes packages by threshold rules such as size and weight.
- Authored the core paper design and personally completed about half the SolidWorks CAD transfer, directing teammates on the remainder from my specifications, reaching 99% sortation accuracy and 20,000 to 30,000 packages per hour while cutting required labour from 20 to 40 workers per shift down to 8.
- Built the financial model behind the automation case, projecting about 200% ROI on machinery investment and 90% higher profit than the current manual design with 28% profit growth in year one.
- Took over client-facing decisions when the client's primary contact abruptly stopped responding mid-project, directing teammates to research alternatives and evaluating feasibility against known requirements. GoBolt has since implemented roughly 70% of the delivered design into their live system.

Euromaxx Logistics Ltd.

Chattogram, Bangladesh

Rotational Intern, Best Intern Award

Mar 2023 – Sep 2023

- Rotated through 4 departments analyzing import and export transaction data and coordinating international shipping and customs documentation, improving on-time delivery to 98% or higher and cutting processing time 20%.
- Managed order tracking for 10 or more international buyers in Salesforce and NetSuite, cutting response time 40% and documentation errors about 30%.

Aarong

Chattogram, Bangladesh

Cashier and Sales Associate, Best Employee of the Year among 70 or more staff

May 2024 – Aug 2024

- Assisted 150 or more customers daily across departments with 100% POS transaction accuracy.
- Averaged about 3,000 dollars in transactions per shift, contributing to being named Best Employee of the Year among 70 or more staff.

PROJECTS

Voyagista Explorer

Toronto, ON

Street Map and Courier Routing Application, C++, EZGL, GTK

Jan 2026 – Apr 2026

- Implemented driving, walking, and walk-to-pickup routing on Dijkstra shortest-path search with one-way restrictions and turn penalties.
- Cut repeated pickup-search computation about 35% by replacing per-candidate forward searches with a single reverse search from the destination.
- Built a multi-threaded courier-delivery optimizer with parallel all-pairs precomputation using `std::async`.
- Implemented iterated local search with double-bridge perturbation and simulated-annealing-style acceptance, coordinating threads with `std::atomic` and `std::mutex`, accelerating preprocessing 2 to 4 times and improving route quality about 20%.

MedFM

Medication Adherence and Safety Platform, Expo, React Native, TypeScript, Node.js, Firebase

Jan 2026 – Present, paused

- Architected a web-first MVP with a unified cross-platform frontend on a serverless Node.js backend, cutting duplicated platform-specific work about 35%.
- Engineered multi-provider authentication, including email and password, Google OAuth, and SMS OTP, with session-state handling across providers.
- Built a 365-day scheduling engine with dynamic event generation and rate-limited, server-validated REST API endpoints.

Embedded Systems and Avionics

UofT Rocketry Association, UTRA

Altium Designer, C, C++, I2C, SPI, UART

Mar 2025 – Present

- Architected multi-layer PCBs integrating microcontrollers, power regulation, and sensor interfaces for rocket telemetry, the board other subteams' sensors plug directly into.
- Programmed embedded C and C++ firmware for real-time sensor acquisition over I2C, SPI, and UART.

- Cut sensor-to-telemetry latency about 25% versus the previous polling approach.

UTMIST AI2

Reinforcement Learning Fighting-Game Agent, Python, OpenAI Gym, Stable Baselines 3, PPO

Nov 2025

- Built a custom Gym environment and PPO agent, defining the observation and action interface and reward signal.
- Improved training convergence and combat behavior about 25% through reward shaping, reaching Top 10 placement and presenting at EXPO.
- Ported the trained model to Tenstorrent's TTNN accelerator stack, adapting tensor operations from PyTorch to a grid-based dataflow architecture.

Voice-Controlled Ball Survival Game

ECE243 Final Project

Assembly, C, memory-mapped IO, VGA graphics, audio input, DE1-SoC

Jan 2026 – Apr 2026

- Built a real-time game in assembly and C on the DE1-SoC using direct memory-mapped IO for VGA frame-buffer rendering with double buffering to eliminate flicker and screen tearing.
- Implemented live audio input, mapping microphone amplitude directly to lift force as the core survival mechanic.
- Detected pitch in real time using a zero-crossing algorithm to drive gameplay difficulty.
- Implemented a finite-state game loop with start, gameplay, and game-over states and a live-updating HUD showing score, survival time, and balls remaining, validated on both the CPULator simulator and physical DE1-SoC hardware.

FPGA Reaction Game

Verilog, DE1-SoC, Quartus, ModelSim

Sep 2025 – Dec 2025

- Designed hardware-only game logic in Verilog, including debouncing, edge detection, and FSM input validation, testing both reaction speed and sustained focus.
- Randomly assigned each of 3 physical buttons a colour prompt at random time intervals up to 7 seconds apart to test sustained focus rather than pure reaction speed.
- Required a different response per colour, two presses for blue, one press for green, and no press for red, and displayed all on-screen text mirrored as an added cognitive challenge.
- Built a scoring and warning FSM awarding points for correct responses and ending the game after repeated mistakes, verified in ModelSim before bringing it up on physical FPGA hardware.

NeuroVision

Hackathon finalist

Eye-Tracking Accessibility System, Python, OpenCV, Dlib, PyAutoGUI

Sep 2024 – Present

- Built hands-free cursor control from webcam eye-tracking and blink detection at 98% accuracy.
- Cut assistive-hardware cost about 90% versus specialized tracking devices, removing the need for dedicated eye-tracking hardware.

Courtyard Sustainable Food Initiative

UofT ESP I

Team Leader

Sep 2024 – Dec 2024

- Directed a 5-member team through 12 or more stakeholder interviews to understand accessibility and climate-resilience needs.
- Converted interview insights into measurable design requirements for an inclusive food court concept.

LEADERSHIP

Mentor, WISE High School Mentorship Program

- Diagnosed each mentee's skill, emotional, and expectations gaps individually, then built resource guides ahead of each session and escalated to faculty specialists when a gap was outside my depth.
- Spent about 5 hours per week per mentee and was trusted with a third mentee through direct referral.

TECHNICAL SKILLS

Languages: C++ with memory management and concurrency using `std::async`, `std::atomic`, and `std::mutex`, Assembly, Verilog, C, Python, TypeScript

Software and ML: Git, Linux, Node.js, RESTful APIs, OpenCV, PyTorch including SE-UNet and Mamba-style models, reinforcement learning using PPO, Tenstorrent TTNN

Hardware and CAD: Altium Designer, I2C, SPI, UART, Quartus, ModelSim, SolidWorks

Cloud and DevOps: Azure DevOps including Repos, Pipelines, and Test Plans, Azure GPU-enabled VMs

Business Tools: Salesforce, NetSuite, financial modeling